

Sanchay Narendra Gawande

sanchay823@gmail.com | 857-395-5752 | Boston, MA | LinkedIn: <https://www.linkedin.com/in/sanchay-gawande/> |
GitHub: <https://github.com/SanchayGawande> | Portfolio: <https://sanchaygawande.com/>

SUMMARY

Software Engineer with 2+ years of experience building scalable backend systems, AI platforms, and cloud native applications. Experienced in Python, JavaScript, AWS, microservices, distributed systems, REST APIs, and GraphQL. Experience building production AI systems, including LLM orchestration platforms, RAG pipelines, and event-driven data architectures supporting real-world AI workflows. Improved system performance by 30% to 50%, reduced latency by 40%, and strengthened reliability through observability, monitoring, and resilient architecture. Passionate about building high-performance backend and AI-driven systems at scale.

TECHNICAL SKILLS

- Programming:** Python, JavaScript, Java, SQL, TypeScript, C, C++, Bash, JSON, YAML
- Backend Systems:** FastAPI, Flask, Node.js, Express, REST APIs, GraphQL, Microservices Architecture, Distributed Systems, API Design, Asynchronous Processing
- Data Systems:** PostgreSQL, MongoDB, Redis, Firebase, Pinecone, Query Optimization, Indexing, Data Modeling, Data Pipelines, Caching
- Frontend:** React, Next.js, Redux, D3.js, HTML5, CSS3
- Cloud and Infrastructure:** AWS Lambda, ECS, SQS, API Gateway, EC2, S3, CloudFront, Docker, Kubernetes, Terraform, CloudWatch, GCP, Azure
- AI Systems:** RAG, LLM Orchestration, LangChain, LangGraph, OpenAI API, Embeddings, DistilBERT, NLP, Model Inference, AI Agents
- Testing:** Pytest, Jest, Cypress, Unit Testing, Integration Testing
- DevOps:** CI/CD, GitHub Actions, Infrastructure as Code, Monitoring, Logging, Deployment Automation
- Engineering:** Data Structures, Algorithms, Object-Oriented Programming, System Design, Event Driven Architecture, Observability, Performance Optimization, Scalability
- Tools:** Git, GitHub, Jira, Postman, Linux, Agile, Scrum, SDLC

WORK EXPERIENCE

Founding Machine Learning Engineer, Suno Analytics

January 2025 to Present | Remote, USA

- Designed and implemented core backend services and ML inference workflows using Python, FastAPI, and PostgreSQL to support an AI-driven ERP analytics platform processing financial and operational datasets.
- Built scalable data ingestion and preprocessing pipelines using AWS Lambda, SQS, and asynchronous processing to prepare enterprise datasets for LLM analysis, reducing ingestion latency by 40%.
- Developed a production retrieval augmented generation system using Pinecone, embeddings, and metadata routing, enabling contextual financial analysis across multi-step workflows.
- Implemented multi-model LLM orchestration integrating OpenAI, Claude, and LLaMA with routing logic based on cost, latency, and task complexity, improving efficiency.
- Improved platform reliability by implementing retry strategies, idempotent processing, validation layers, and failure isolation, reducing pipeline failures by 45%.
- Established observability across backend and ML services using AWS CloudWatch, structured logging, and alerting reducing debugging and incident resolution time by 35%.
- Optimized database queries, API performance, and inference execution using indexing, caching, and background processing, maintaining P99 latency under 200 ms and reducing report generation time by 30%.
- Built internal evaluation tooling and structured data models, translating complex financial workflows into ML-ready representations, enabling reliable automation and decision support.

Software Engineer Fellowship, Headstarter

July 2024 to January 2025 | Remote, USA

- Delivered multiple full-stack and AI-driven applications end-to-end using Python, FastAPI, Node.js, React, PostgreSQL, and AWS, shipping production-quality features under Agile weekly release cycles managed in Jira.
- Integrated AI/ML features, including LLM-powered components, inference workflows, and data processing services into application architectures to ensure stable and predictable execution.
- Led peer code reviews, provided PR-level feedback, and enforced coding practices that improved code quality and reduced debugging time across the cohort.
- Created internal utilities, debugging scripts, and deployment helpers that accelerated development velocity for multi-week team projects.
- Completed recurring engineering challenges with strict deadlines that replicated real production sprints, consistently ranking as a top performer in the fellowship.
- Shipped 8 production-style releases across 5 full-stack applications, delivering new features every week while maintaining consistent build and deploy hygiene.
- Set up CI/CD automation using GitHub Actions, reducing deployment time by 50% and cutting recurring integration issues by 30% through repeatable checks and standardized workflows.
- Reviewed 40+ pull requests and drove consistent code standards, reducing rework cycles by 25% and accelerating peer delivery during weekly sprints.

Software Engineer Intern, University of Massachusetts Boston

January 2024 to May 2024 | Boston, USA

- Architected a decoupled patient guidance platform, separating the React frontend from the Python inference backend to enable independent scaling and faster iteration.
- Implemented a hybrid search engine combining vector retrieval with metadata rules to structure nursing content, reducing LLM hallucinations in outputs.
- Collaborated in a 4-person cross-functional team to ship iConcern, a portal for diabetes-related patient education and support, aligning backend services with frontend and ML workflows.
- Owned the backend authentication track, implementing secure sign-in and session-handling patterns to protect user data and enable 3 core capabilities: educational articles, personalized chat assistant, and secure authentication.

PROJECTS

PeerGenius - AI-powered student collaboration platform

- Deployed a context-aware AI response system with 95% accuracy for academic query detection, differentiation between solo and group study modes, and Firebase Authentication with JWT security.
- Developed a full-stack architecture with React 19, Express, MongoDB, and a CI/CD pipeline with production security standards.

LifeLens - Microservices-based AI daily decision engine

- Built a full-stack microservices platform with Node.js, TypeScript, React Native, and FastAPI, integrating 5+ AI APIs with fallback mechanisms and 95%+ uptime.
- Optimized PostgreSQL (Row Level Security, indexing) and Redis caching, boosting performance by 30% to 50% via token optimization and cache invalidation while keeping query times under 200 ms.

EDUCATION

University of Massachusetts Boston | Boston, MA, USA

Master of Science in Computer Science

Aug 2022 to May 2024

Sant Gadge Baba Amravati University | Maharashtra, India

Bachelor of Engineering in Computer Science and Engineering

June 2019 to May 2022